News Article Entity Extraction

Part A (Coding): Clean this list of articles (duplicate titles, inconsistent date formats) and compute how many articles were published per topic per calendar week.

[

{"title": "City Council Approves New Transit Line", "topic": "local", "date": "2026-08-01"},

{"title": "City Council Approves New Transit Line", "topic": "Local", "date": "08/01/2026"},

{"title": "Tech Firm Announces Layoffs", "topic": "business", "date": "2026-08-03"},

{"title": "Regional Team Wins Championship", "topic": "sports", "date": "2026-08-06"},

{"title": "Tech Firm Opens New Office", "topic": "business", "date": "2026-08-09"}

]

Part B (Evals + Knowledge Graph):

1. **Evaluate**: Below is an LLM's extraction of people/organizations from a sentence vs. the reference extraction. Score precision/recall and explain which errors matter more.
    - Text: *"Maria Gomez, CEO of Northwind Logistics, announced a partnership with the city of Denver."*
    - Reference: People = [Maria Gomez]; Organizations = [Northwind Logistics]; Locations = [Denver]
    - LLM output: People = [Maria Gomez]; Organizations = [Northwind Logistics, Denver]; Locations = []
2. **Model as a graph**: Build a graph with edges like Person —worksAt→ Organization —locatedIn→ Location, and Article —mentions→ Person/Organization. Write a query that finds all organizations mentioned in the same article as a given person.
3. **Share github link of this assignment**